Ryan Sheatsley

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Academic & Professional Experience

United States Army Research Laboratory — **Adelphi, MD**
Research Scientist and Team Lead (mentored by Andrew Toth) — *Summers of 2014–2018*

Patent Litigation — **University Park, PA**
Code Review & Document Drafting — *2018–2019*

Systems Internet and Infrastructure Security Laboratory — **University Park, PA**
Systems Administrator — *2015–Present*

Penn State Learning Center — **University Park, PA**
Writing Tutor (mentored by Dr. Jon Olson) — *2012–2013*

Education

Pennsylvania State University — **University Park, PA**
M.S. in Computer Science and Engineering — *2016–2018*

- Thesis: *Adversarial Examples in Constrained Domains*
- Advisor: *Prof. Patrick McDaniel*
- Thesis committee: Prof. Patrick McDaniel, Prof. Trent Jaeger, Prof. Nicolas Papernot

Pennsylvania State University — **University Park, PA**
B.S. in Computer Engineering — *2011–2015*

Publications

Conference Proceedings

Characterizing the Modification Space of Signature IDS Rules. *Ryan Guide, Eric Pauley, Yohan Beugin, Ryan Sheatsley, Patrick McDaniel*. Proceedings of the 2023 Military Communications Conference (MILCOM), Boston, MA. (2023)

The Space of Adversarial Strategies. *Ryan Sheatsley, Blaine Hoak, Eric Pauley, Patrick McDaniel*. Proceedings of the 32nd USENIX Security Symposium (USENIX), Anaheim, CA. (2023)

Measuring and Mitigating the Risk of IP Reuse on Public Cloud. *Eric Pauley, Ryan Sheatsley, Blaine Hoak, Quinn Burke, Yohan Beugin, Patrick McDaniel*. Proceedings of the 43rd IEEE Symposium on Security and Privacy (S&P), San Francisco, CA. (2022)

A Machine Learning and Computer Vision Approach to Geomagnetic Storm Forecasting. *Kyle Domico, Ryan Sheatsley, Yohan Beugin, Quinn Burke, Michael Patrick McDaniel*. Proceedings of the Second AGU Conference on Machine Learning in Heliophysics (ML-Helio), Boulder, CO. (2022)

Building a Privacy-Preserving Smart Camera System. *Yohan Beugin, Quinn Burke, Blaine Hoak, Ryan Sheatsley, Eric Pauley, Gang Tan, Syed Rafiul Hussain, Patrick McDaniel*. Proceedings on Privacy Enhancing Technologies Symposium (PETS), Sydney, Australia. (2022)

HoneyModels: Machine Learning Honeypots. *Ahmed Abdou, Ryan Sheatsley, Yohan Beugin, Tyler Shipp, Patrick McDaniel*. Proceedings of the 2021 Military Communications Conference (MILCOM), San Diego, CA. (2021)
On the Robustness of Domain Constraints. *Ryan Sheatsley, Blaine Hoak, Eric Pauley, Yohan Beguin, Yohan Beugin, Michael Weisman, Patrick McDaniel*. Proceedings of the 2021 Conference on Computer and Communications Security (CCS), Coex, Seoul, Republic of Korea. (2021)
Feature Engineering: A Case Study For Radiation Source Localization In Complicated Environments. *Matthew Durbin, Ryan Sheatsley, Patrick McDaniel, Azaree Lintereur*. Proceedings of the 62nd Annual Meeting of the Institute of Nuclear Materials Management (INMM), Virtual. (2021)
A Multi-Step Machine Learning Approach to Directional Gamma Ray Detection. *Matthew Durbin, Ryan Sheatsley, Patrick McDaniel, Azaree Lintereur*. Proceedings of the 61st Annual Meeting of the Institute of Nuclear Materials Management (INMM), Virtual. (2020)
Development of Machine Learning Algorithms for Directional Gamma Ray Detection. *Matthew Durbin, Ryan Sheatsley, Christopher Balbier, Tristan Grieve, Patrick McDaniel, Azaree Lintereur*. Proceedings of the 60th Annual Meeting of the Institute of Nuclear Materials Management (INMM), Palm Springs, CA. (2019)
Curie: Policy-based secure data exchange. *Z. Berkay Celik, Abbas Acar, Hidayet Aksu, Ryan Sheatsley, Patrick McDaniel, A Secuk Uluagac*. Proceedings of the Ninth ACM Conference on Data and Application Security and Privacy (CODASPY), Richardson, TX. (2019)
Network Traffic Obfuscation: An Adversarial Machine Learning Approach. *Gunjan Verma, Ertugrul Ciftcioglu, Ryan Sheatsley, Kevin Chain, Lisa Scott*. Proceedings of the 2018 Military Communications Conference (MILCOM), Baltimore, MD. (2018)
Detection under Privileged Information. *Z. Berkay Celik, Patrick McDaniel, Rauf Izmailov, Nicolas Papernot, Ryan Sheatsley, Raquel Alvarez, Ananthram Swami*. Proceedings of the 2018 Asia Conference on Computer and Communications Security (ASIACCS), Incheon, Republic of Korea. (2018)
Heterogeneous information sharing of sensor information in contested environments. *Jason A Wampler, Chien Hsieh, Andrew Toth, Ryan Sheatsley*. Proceedings of Volume 10190 Ground/Air Multisensor Interoperability, Integration, and Networking for Persistent ISR VIII (SPIE), Anaheim, CA. (2017)

Journals

Experimental tests of Gamma-ray Localization Aided with Machine-learning (GLAM) capabilities. *Matthew Durbin, Ryan Sheatsley, Patrick McDaniel, Azaree Lintereur*. Elsevier Journal of Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment (NIM-A). (2022)
Adversarial Examples for Network Intrusion Detection Systems. *Ryan Sheatsley, Nicolas Papernot, Michael Weisman, Gunjan Verma, Patrick McDaniel*. IOS Press Journal of Computer Security (JCS). (2022)
Physics-based Misbehavior Detection System for V2X Communications. *Alejandro Andrade Salazar, Patrick McDaniel, Ryan Sheatsley, Jonathan Petit*. SAE International Journal of Connected and Automated Vehicles. (2021)
Improving Radioactive Material Localization by Leveraging Cyber-Security Model Optimizations. *Ryan Sheatsley, Matthew Durbin, Azaree Lintereur, Patrick McDaniel*. IEEE Sensors. (2021)

Book Chapters

Evading Machine Learning-based Network Intrusion Detection Systems with GANs. *Bolor-Erdene Zolbayar, Ryan Sheatsley, Patrick McDaniel*. In: *Charles A Kamhoua, Christopher D. Kiekintveld, Fei Fang, Quanyan Zhu* (Ed.). Game Theory and Machine Learning for Cyber Security. John Wiley & Sons, Incorporated, Hoboken, New Jersey. (2021)

Technical Reports

Systematic Evaluation of Geolocation Privacy Mechanisms. *Alban Héon, Ryan Sheatsley, Quinn Burke, Blaine Hoak, Eric Pauley, Yohan Beugin, Patrick McDaniel*. (2023)
Generating Practical Adversarial Network Traffic Flows using NIDSGAN. *Bolor-Erdene Zolbayar, Ryan Sheatsley, Patrick McDaniel, Michael Weisman, Sencun Zhu, Shitong Zhu, Srikanth Krishnamurthy*. (2020)
Adversarial Planning. *Valentin Vie, Ryan Sheatsley, Sophia Beyda, Sushrut Shringarputale, Kevin Chan, Trent Jaeger, Patrick McDaniel*. (2020)
A Vision Toward an Internet of Battlefield Things (IoBT): Autonomous Classifying Sensor Network. *John Zhu, Egan McClave, Quan Pham, Sujay Polineni, Sam Reinhart, Ryan Sheatsley, Andrew Toth*. (2018)
Cleverhans v1.0.0: an adversarial machine learning library. *Nicolas Papernot, Ian Goodfellow, Ryan Sheatsley, Reuben Feinman, Patrick McDaniel*. (2016)
Analyzing GAIAN Database (GaianDB) on a Tactical Network. *Ryan Sheatsley, Andrew Toth*. (2015)

Thesis

Adversarial Examples in Constrained Domains. *Ryan Sheatsley*. (2018)

Honors

Robert M. Owens Memorial Scholarship: Pennsylvania State University *2017*
Pathways Internship Program: United States Government *2015*
Peer Tutor Faculty Recommendation: Pennsylvania State University *2011*

Talks

On the Robustness of Domain Constraints: CTML Industrial Advisory Board Meeting *2021*
Adversarial Examples in Constrained Domains: RIT Great Lakes Security Day *2020*
Adversarial Examples in Constrained Domains: Cyber-CRA Webinar *2020*

Professional Activities

Reviewer

SATML: IEEE Symposium on Secure and Trustworthy Machine Learning *2023*
IEEE S&P: IEEE Symposium on Security and Privacy *2022*

JCS: IOS Press Journal of Computer Security *2022*
SOFTW: IET Software *2022*
TDSC: IEEE Transactions on Dependable and Secure Computing *2021*
IC: IEEE Internet Computing *2021*
TIP: IEEE Transactions on Image Processing *2020*
TEM: IEEE Transactions on Engineering Management *2020*
DKE: Data & Knowledge Engineering Journal *2020*
GLOBECOM: IEEE Global Communications Conference *2019*

Volunteer

Dancing with Robots Penn State Summer Camp Coordinator *2019*